

Research Papers Report

Robust Reinforcement Learning

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Executive Summary

Robust Reinforcement Learning (Robust RL) addresses one of the most critical open challenges in deploying RL agents: ensuring policies remain effective under distribution shifts, adversarial perturbations, model uncertainty, and corrupted feedback. The six papers reviewed span the core dimensions of this challenge—from adversarial attacks on state observations, distributionally robust MDPs, adversarial zero-sum games, offline RL under transition uncertainty, risk-sensitive CVaR optimization, to robustness against corrupted human preferences.

Together these works reveal a rich ecosystem: **SA-MDP/RADIAL** (NeurIPS 2020) establishes a principled foundation for observation-space robustness; **QARL** (ICLR 2024) reformulates adversarial RL as a curriculum over bounded-rationality agents; **OPROVI-TV** (NeurIPS 2024) closes the sample-complexity gap for distributionally robust RL with interactive data; **RRPI** (arXiv 2026) extends offline RL to worst-case transition dynamics via KL-regularized policy iteration; **Robust CVaR RL** (2024) bridges risk-sensitive and robust RL through NCVaR; and **R³M** (NeurIPS 2024) introduces robust RLHF handling corrupted human preference labels.

Summary Comparison Table

Title (Short)	Year	Key Contribution	Best Result	Main Limitation
SA-MDP / RADIAL (Zhang et al.)	NeurIPS 2020	State-Adversarial MDP + policy regularizer for obs-space robustness	Strong robustness on 10 envs; action guarantees on Pong	Stationary optimal policy may not exist; only obs perturbations
QARL (Reddi et al.)	ICLR 2024	Bounded-rationality curriculum via Quantal Response Equilibrium	Outperforms RARL on DMControl locomotion & navigation	Curriculum schedule is heuristic; two-player games only
OPROVI-TV (Lu et al.)	NeurIPS 2024	Near-optimal interactive DR-RL; establishes curse-of-support-shift hardness	Provably near-optimal sample complexity for TV uncertainty sets	VMV assumption restricts scope; tabular setting only
RRPI	arXiv 2026	KL-regularized robust policy iteration for offline RL	SOTA on D4RL benchmarks vs. pessimistic offline RL baselines	KL sets may be restrictive; no deep RL theory yet
Robust CVaR RL (Ni & Lai)	arXiv 2024	NCVaR measure linking CVaR risk-sensitivity to robust MDP	Convergent value iteration with equivalence theorem	Theoretical focus; limited empirical benchmarking

R³M (Bukharin et al.)	NeurIPS 2024	■-regularized MLE for robust RLHF under corrupted preferences	15-40% gain over standard RLHF; 5-12pp over DPO on NLG	Assumes sparse corruption; Bradley-Terry model only
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Detailed Paper Reviews

Paper 1

■ Robust Deep Reinforcement Learning against Adversarial Perturbations on State Observations

Authors: Huan Zhang, Hongge Chen, Chaowei Xiao, Bo Li, Mingyan Liu, Duane Boning, Cho-Jui Hsieh
Year / Venue: 2020 — NeurIPS 2020 (Spotlight)
Source: <https://arxiv.org/abs/2003.08938>

■ Problem Statement

DRL agents observe states through potentially noisy or adversarially-perturbed observations. Small perturbations (e.g., sensor noise, adversarial pixel attacks) can mislead an agent into catastrophically suboptimal actions. Naively borrowing adversarial training from computer vision was ineffective for RL. No rigorous theoretical framework existed for characterizing robustness in sequential decision-making.

■ Proposed Method

State-Adversarial MDP (SA-MDP): The authors define an SA-MDP where an adversary perturbs the agent's observation within an ϵ ball of radius ϵ before action selection. Key findings: (a) a stationary Markovian optimal policy may not exist in SA-MDP; (b) the optimal adversarial strategy is state-dependent and potentially non-stationary.

Policy Regularization: A principled robust regularizer penalizes the policy for producing different actions under perturbed observations, using either KL-divergence or total variation between clean and adversarially-perturbed action distributions. Added to PPO, DDPG, and DQN objectives. Worst-case perturbation is found via PGD.

Novel Attacks: Two strong white-box attacks are introduced: Robust Sarsa (RS) attack (Q-function sensitivity) and Maximal Action Difference (MAD) attack (maximizes KL between clean/perturbed actions).

■ Comparison with Previous Methods

Prior adversarial training from image classification (FGSM, PGD) could not account for temporal RL dynamics—policies overfit to perturbed training distribution but remained brittle. The SA-MDP framework provides the first theoretically grounded formulation, and the regularizer avoids repeated expensive inner attack computations compared to naive adversarial training.

■ Framework Architecture

SA-MDP loop: environment transitions to $s_t \rightarrow$ adversary generates $\tilde{s}_t = s_t + \delta$ (bounded by ϵ) $\rightarrow$ agent selects a_t from $\tilde{s}_t$. Training adds robustness regularizer: $L_{\text{robust}} = L_{\text{original}} + \lambda \cdot D(\pi(\cdot|s), \pi(\cdot|\tilde{s}))$, alternating between (1) PGD-style attack to find worst-case δ and (2) policy update minimizing both task objective and regularizer.

■ Experimental Results

Environments: 10 environments — 6 Atari games (discrete) + 4 MuJoCo tasks (continuous).

Results: RADIAL-PPO/DDPG/DQN significantly outperforms vanilla baselines under RS and MAD attacks. On HalfCheetah, unprotected DDPG scores near 0 under attack; RADIAL-DDPG retains >70% of clean performance. On Pong, formal action-consistency guarantee is provided for all rollout frames.

Clean Performance: Robust training also improves clean-environment performance in several tasks, suggesting the regularizer acts as a beneficial smoothness constraint.

■ Critical Feedback

- SA-MDP theory shows no stationary optimal policy exists, yet a stationary policy is still trained—this gap is unresolved.
- Only ϵ -bounded observation perturbations considered; action/reward/dynamics perturbations out of scope.
- PGD inner attack during training is computationally expensive; scalability is limited.
- No evaluation against adaptive or black-box attacks.

■ Robust Adversarial Reinforcement Learning via Bounded Rationality Curricula (QARL)

Authors: Aryaman Reddi, Maximilian Tölle, Jan Peters, Georgia Chalvatzaki, Carlo D'Eramo

Year / Venue: 2023 / ICLR 2024 — ICLR 2024

Source: <https://arxiv.org/abs/2311.01642>

■ Problem Statement

RARL trains a protagonist against an adversary in a zero-sum Markov game. The optimal solution is a Nash Equilibrium (NE), but NE computation requires solving a non-convex saddle-point optimization—intractable for high-dimensional continuous control. In practice, RARL training is highly unstable and often diverges: the adversary quickly destabilizes the protagonist before it learns any useful behavior.

■ Proposed Method

Entropy-Regularized Markov Game & QRE: Entropy regularization is added to both players' objectives, transforming the hard NE problem into finding a Quantal Response Equilibrium (QRE). In QRE, agents play sub-optimally with probability proportional to action values (bounded rationality). The temperature β controls rationality: $\beta \rightarrow \infty$ recovers NE, $\beta \rightarrow 0$ produces random play.

QARL Curriculum: β is gradually increased from small (irrational adversary) to large (fully rational adversary). Early training: stable learning as the weak adversary allows basic skill acquisition. Late training: increasingly strategic adversary pushes policy toward robustness. Both agents use SAC-style updates.

Adversary: Applies bounded additive force perturbations to the protagonist's action/dynamics (e.g., joint torques in MuJoCo). Training alternates protagonist and adversary updates.

■ Comparison with Previous Methods

■ Framework Architecture

Two-player competitive loop: protagonist π_θ and adversary π_ϕ interact with the environment simultaneously. Adversary applies bounded perturbations to protagonist's dynamics. Both use SAC updates with temperature β_t . Curriculum scheduler increases β_t (linear/exponential annealing) over training. QRE adversary objective: $\max_\phi E[Q^{\text{adv}}(s,a)] - \beta_t \cdot H(\pi_\phi)$.

■ Experimental Results

Environments: DeepMind Control Suite — HalfCheetah, Hopper, Walker, Ant (locomotion); PointMass, Reacher (navigation).

Results: QARL outperforms RARL across all 6 environments in both nominal and adversarial conditions. On HalfCheetah-Hard: QARL ~800 vs. RARL ~200. RARL catastrophically fails on PointMass navigation; QARL succeeds.

Ablations: Constant high- $\beta \rightarrow$ RARL-like instability. Constant low- $\beta \rightarrow$ weak policy failing under adversarial attack. Curriculum is essential.

■ Critical Feedback

- Curriculum annealing schedule is a sensitive hyperparameter with no automatic tuning procedure.
- Limited to two-player competitive games; multi-agent extensions not addressed.
- Adversary models action/dynamics perturbations; observation-space adversaries not considered.
- Convergence guarantees for the QRE curriculum approach not fully established.

■ Distributionally Robust RL with Interactive Data Collection: Fundamental Hardness and Near-Optimal Algorithm

Authors: Miao Lu, Han Zhong, Tong Zhang, Jose Blanchet

Year / Venue: 2024 — NeurIPS 2024

Source: <https://arxiv.org/abs/2404.03578>

■ Problem Statement

DR-RL trains policies optimal under worst-case transition dynamics within an uncertainty set (RMDP), addressing the sim-to-real gap. All prior DR-RL work required a generative model (perfect simulation oracle) or an offline dataset with good deployment coverage—both unrealistic. This is the first work to study DR-RL in the fully interactive online setting, where the learner only interacts with the training environment.

■ Proposed Method

Hardness Result (Curse of Support Shift): The authors prove that without additional assumptions, no algorithm achieves sample-efficient learning in interactive DR-RL. Training and test environments may have disjoint state-action support, making value estimation impossible in unseen regions.

Vanishing Minimal Value (VMV) Assumption: The minimum robust value function value over all states equals zero. Naturally satisfied in episodic MDPs where agents must reach a terminal state. Under VMV, support shift is effectively eliminated for TV-distance uncertainty sets.

OPROVI-TV Algorithm: Uses optimism (UCB exploration) for the training environment and pessimism for worst-case test environment evaluation. Confidence sets are built from empirical transition concentration inequalities. Sample complexity: $\tilde{O}(S^2A/((1-\gamma)^3\epsilon^2))$ —nearly matching the information-theoretic lower bound.

■ Comparison with Previous Methods

All prior DR-RL works (Yang et al. 2021, Shi & Chi 2022, Panaganti & Kalathil 2022) required generative models or well-covered offline data. OPROVI-TV is the first with provable guarantees for online interactive DR-RL and improves sample complexity over offline methods that required $O(S^3)$ due to redundant state coverage.

■ Framework Architecture

Epoch-based algorithm: (1) interact with training MDP M_0 using UCB exploration, collect transitions; (2) build UCB estimates $P_UCB(s'|s,a)$; (3) apply pessimistic robust Bellman operator: $V_k(s) = \min_{P' \in B_TV(P, \rho)} \sum_{s'} P'(s'|s, \pi(s)) \cdot V_k(s')$; (4) extract robust policy greedy w.r.t. V_k ; repeat until convergence.

■ Experimental Results

Focus: Primarily theoretical. Synthetic tabular grid-world experiments confirm that OPROVI-TV achieves robust optimality with fewer samples than naive online RL. Experiments validate the necessity of the VMV assumption by demonstrating failure without it.

Sample Complexity: $\tilde{O}(S^2A/((1-\gamma)^3\epsilon^2))$ for ϵ -optimal robust policy. Matches lower bound up to logarithmic factors.

■ Critical Feedback

- VMV assumption may not hold in all practical problems; requires verification.
- Polynomial in S and A ; not directly applicable to continuous state-action spaces.
- Only TV-distance uncertainty sets covered; KL and chi-squared sets are future work.

- No empirical validation on standard deep RL benchmarks (MuJoCo, Atari).

■ Robust Regularized Policy Iteration under Transition Uncertainty (RRPI)

Authors: Anonymous (under review)
Year / Venue: 2026 — arXiv:2603.09344 (March 2026)
Source: <https://arxiv.org/abs/2603.09344>

■ Problem Statement

Offline RL learns from fixed datasets without further interaction. Distributional shift causes agents to visit OOD state-action pairs with unreliable value estimates. Standard methods (pessimism, behavioral constraints) do not explicitly model transition uncertainty—if the learned dynamics are inaccurate, the policy may exploit these inaccuracies. RRPI formulates offline RL as a robust MDP problem over a KL-distance uncertainty set around the empirical transition kernel.

■ Proposed Method

Robust Policy Optimization: $\max_{\pi} \min_{P \in \mathcal{U}} V^{\pi}_P$. The max-min bilevel problem is intractable for general uncertainty sets.

KL-Regularized Surrogate: Replace min with KL regularizer: $\max_{\pi} E_{P^{\square}}[V^{\pi}] - \lambda \cdot \text{KL}(P || P^{\square})$. The inner minimization has a closed-form solution via exponential tilting, making the surrogate tractable. The surrogate lower-bounds the robust value under KL-uncertainty sets.

Robust Regularized Bellman Operator: T_{λ} is proven to be a γ -contraction in L^{∞} . Policy iteration via T_{λ} converges to unique fixed point V^*_{λ} . Monotone improvement of the robust objective is guaranteed throughout training.

■ Comparison with Previous Methods

vs. IQL/TD3+BC/CQL: RRPI explicitly models transition uncertainty, not just policy constraints. vs. MOPO/COMBO: No explicit dynamics model required—KL uncertainty set is directly parameterized from data. vs. ATAC/PRDC: RRPI provides monotone improvement guarantees for the robust objective.

■ Framework Architecture

Pipeline: (1) estimate $P^{\square}(s'|s,a)$ from offline dataset D via MLE; (2) construct T_{λ} : $[T_{\lambda}V](s,a) = r(s,a) + \gamma E_{P^{\square}}[V(s')] - \lambda \cdot \log Z(s,a,V/\lambda)$, where Z is the log-partition function; (3) iterate policy evaluation via T_{λ} until convergence to V^*_{λ} ; (4) extract greedy policy; (5) repeat (policy iteration). No adversarial training loops required.

■ Experimental Results

Benchmark: D4RL — MuJoCo (HalfCheetah, Hopper, Walker2d) across medium, medium-replay, medium-expert datasets.

Results: RRPI achieves strong normalized scores (~79 on Hopper-medium-expert, ~87 on Walker2d-medium-expert), outperforming percentile-based offline RL methods on most environments. Strongest gains on medium-replay datasets where distributional shift is most severe.

Robustness Check: Lower Q-values align with high epistemic uncertainty (ensemble disagreement), confirming that robust uncertainty sets prevent OOD exploitation.

■ Critical Feedback

- KL uncertainty sets may be too restrictive for large distribution shifts; TV/Wasserstein could be more appropriate.
- λ requires manual tuning; no automatic selection procedure provided.

- Theory limited to tabular/linear approximation; no deep RL guarantees.
- Very recent (March 2026), not yet peer-reviewed or externally reproduced.

■ Robust Risk-Sensitive Reinforcement Learning with Conditional Value-at-Risk

Authors: Xinyi Ni, Lifeng Lai

Year / Venue: 2024 — arXiv:2405.01718 (May 2024)

Source: <https://arxiv.org/abs/2405.01718>

■ Problem Statement

Risk-sensitive RL optimizes CVaR of the return distribution, ensuring worst-case performance under intrinsic MDP randomness. Robust RL optimizes under worst-case transition dynamics (model uncertainty). These two paradigms were studied independently. This paper establishes a formal theoretical equivalence, enabling risk-sensitive RL techniques to directly solve robust RL problems.

■ Proposed Method

NCVaR — Normalized CVaR: $\text{NCVaR}_\alpha(X) = \text{CVaR}_\alpha(X) / (\alpha \cdot E[X])$. Scale-invariant and coherent, with a natural connection to robust optimization duality.

Equivalence Theorem: $\max_\pi \text{NCVaR}_\alpha(V^\pi) \equiv \max_\pi \min_{P \in U_{\text{CVaR}}} E_P[V^\pi]$, where U_{CVaR} is a specific CVaR-induced transition uncertainty set. Optimizing NCVaR is equivalent to solving a robust MDP, bridging risk-sensitive and robust RL communities.

NCVaR Value Iteration: Modified Bellman operator incorporating CVaR over next-state value distribution. Proven contraction guaranteeing convergence. Implementable as dynamic programming without explicit uncertainty set optimization.

■ Comparison with Previous Methods

Prior CVaR-RL (Tamar 2015, Chow & Ghavamzadeh 2014) optimized CVaR directly without robustness connection. Prior robust RL with CVaR uncertainty sets required expensive min-max problems. RRRL (2022) used coherent risk measures without a tractable RL algorithm. This paper provides both equivalence and efficient algorithm.

■ Framework Architecture

Dual equivalent views: (Risk-sensitive) agent maximizes NCVaR_α of return—CVaR tail optimization focuses policy on worst-case outcomes under stochastic dynamics. (Robust) agent solves max-min game against adversary picking worst-case kernel from U_{CVaR} . Equivalence theorem proves identical optimal policies. Value iteration operates in risk-sensitive view for computational efficiency.

■ Experimental Results

Focus: Primarily theoretical. Small-scale MDP experiments (gridworld, inventory control) validate equivalence and convergence.

Results: NCVaR value iteration converges to the optimal robust-risk-sensitive policy. Equivalence theorem numerically confirmed: NCVaR-optimal policies match explicit robust MDP solutions with order-of-magnitude speed advantage.

Risk Ablation: As α decreases (more risk-averse), policies become more conservative and perform better in worst-case scenarios at cost of lower mean performance—confirming theoretical predictions.

■ Critical Feedback

- Primarily theoretical with very limited empirical validation (no MuJoCo, Atari, or real-world tasks).
- NCVaR normalization by $E[X]$ is difficult to estimate online; may introduce bias.

- Extension to deep RL and continuous action spaces not addressed.
- CVaR-induced uncertainty set U_{CVaR} may not align with practical sim-to-real transfer needs.

■ Robust Reinforcement Learning from Corrupted Human Feedback (R³M)

Authors: Alexander Bukharin, Y. Li, Y. Yu, Q. Zhang, Z. Chen, S. Zuo, C. Zhang, S. Zhang, T. Zhao

Year / Venue: 2024 — NeurIPS 2024

Source: <https://arxiv.org/abs/2406.15568>

■ Problem Statement

RLHF aligns LLMs and robotic policies with human preferences, but human annotators are imperfect—they may provide incorrect labels due to bias, fatigue, ambiguity, or adversarial intent. Even a small fraction of corrupted labels can significantly degrade the learned reward model and downstream policy. Standard RLHF has no mechanism to detect or correct such corruptions. R³M is the first robust RLHF framework with both theoretical consistency guarantees and practical efficiency.

■ Proposed Method

Sparse Outlier Model: Corrupted labels are modeled as sparse outliers in the Bradley-Terry preference model. Each comparison has a latent corruption indicator $z_i \in \{0,1\}$: $z_i=0$ clean, $z_i=1$ corrupted. Corruption scales sublinearly: $\sum z_i = o(n)$.

■-Regularized MLE: Reward learning is: $\max_{\theta} \sum_i \log P_{\theta}(y_i | x_i, x_{\blacksquare_i}) - \lambda \|z\|_{\blacksquare}$. The $\blacksquare$ penalty on outlier indicators z promotes sparsity, down-weighting true corruptions.

Alternating Optimization: (1) Reward update: fix z , update θ via gradient ascent on clean-label MLE; (2) Outlier update: fix θ , update z via soft-thresholding ($\blacksquare$ proximal operator). Negligible computational overhead vs. standard RLHF.

■ Comparison with Previous Methods

Standard RLHF (Christiano 2017, Stiennon 2020) and DPO (Rafailov 2023) treat all labels equally with no corruption handling. Prior robust reward learning was computationally expensive and lacked guarantees. R³M provides: formal statistical corruption model, consistency guarantees (ground-truth recovery as $n \rightarrow \infty$), and practical efficiency.

■ Framework Architecture

Pipeline: (1) Human annotators provide preference comparisons $(x_i, x_{\blacksquare_i})$ with potentially corrupted labels y_i ; (2) R³M jointly estimates θ and z via alternating $\blacksquare$ -regularized optimization; (3) Clean labels ($z_i \approx 0$) contribute fully; corrupted labels ($z_i \approx 1$) are down-weighted; (4) Robust reward r_{θ} used in PPO or DPO for policy optimization. Corruption diagnosis module outputs estimated corrupted label set for annotation quality feedback.

■ Experimental Results

Robotic Control: MuJoCo tasks (HalfCheetah, Hopper, Walker) under 3 corruption models—irrational flipping (random), stochastic flipping (probabilistic), myopic flipping (immediate-reward-based). R³M outperforms standard RLHF by 15-40% in normalized reward across all tasks and corruption models.

NLG Tasks: R³M-DPO outperforms vanilla DPO on dialogue (Anthropic HH) and summarization (TL;DR) when 10-30% of preferences are corrupted, with win rates improving by 5-12 percentage points.

Outlier Detection: R³M correctly identifies 85-95% of corrupted labels across corruption rates from 5% to 30%.

■ Critical Feedback

- Sparse corruption assumption fails when corruption is systematic (e.g., persistently biased annotators).
- λ regularization requires tuning; no data-driven selection procedure.
- Complex adversarial annotation strategies (e.g., sycophantic labeling) not evaluated.
- Limited to Bradley-Terry preference model; extensions to cardinal ratings or rankings not addressed.

Research Trends and Future Directions

1. Unification of Robustness Frameworks

The six papers address adversarial, distributional, risk-sensitive, and feedback robustness independently. A unified framework handling all sources of uncertainty—noisy observations, dynamics distribution shift, and corrupted reward signals—simultaneously remains an open challenge.

2. Scalability to Deep RL

The strongest theoretical results (OPROVI-TV, Robust CVaR RL) are limited to tabular settings. Extending distributional robustness guarantees to deep neural network function approximation in continuous state-action spaces is a critical open problem.

3. Robust RLHF for Foundation Models

With RLHF now central to training LLMs (GPT-4, Claude, Gemini), robustness of reward learning to noisy human feedback is of enormous practical importance. R³M is an early step; future work should address systematic annotator biases, reward hacking, and preference inconsistency at scale.

4. Certified Robustness

While several papers provide robustness against specific attack types, formal certification—provably correct behavior under all perturbations within a defined budget—remains elusive for deep RL. Promising directions include interval bound propagation, abstract interpretation, and Lipschitz-constrained neural network policies.